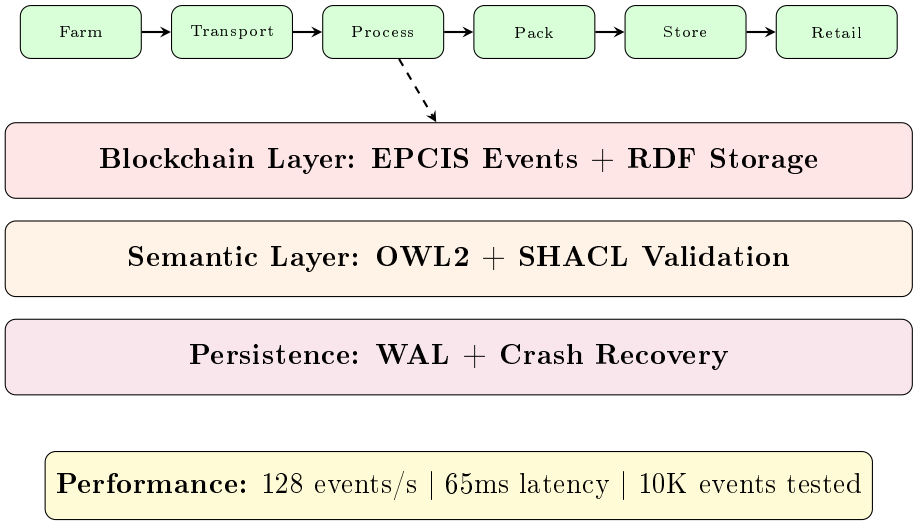
Graphical Abstract

ProvChain: A Generalized Semantic Blockchain Framework for Traceability

Anusorn Chaikaew, Varin Chouvatut, Ekkarat Boonchieng

ProvChain: EPCIS-Compliant Blockchain for UHT Milk Traceability



Performance: 128 events/s | 65ms latency | 10K events tested

GS1 EPCIS 2.0 | Multi-Domain | Healthcare | Pharma | Food | Rust

Highlights

ProvChain: A Generalized Semantic Blockchain Framework for Traceability

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- Generalized blockchain framework integrating GS1 EPCIS 2.0 with semantic web technologies for multi-domain traceability
- Modular ontology architecture supporting food, pharmaceutical, healthcare, and automotive domains
- Domain-specific ontology extension demonstrated through UHT milk case study with SHACL validation
- Write-Ahead Logging (WAL) persistence mechanism with atomic writes and automatic crash recovery
- Interactive demonstration validated across 8 supply chain phases with 7 EPCIS event types
- OWL2 reasoning support for hasKey validation, property chains, and qualified cardinality constraints
- Scalability validated up to 10,000 events with 128 events/second throughput

ProvChain: A Generalized Semantic Blockchain Framework for Traceability

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Abstract

Modern supply chains across diverse industries—including food, pharmaceuticals, healthcare, and manufacturing—face persistent challenges in ensuring product integrity, maintaining quality throughout distribution, and providing end-to-end traceability. This paper presents **ProvChain**, a generalized blockchain framework that integrates GS1 Electronic Product Code Information Services (EPCIS) 2.0 standards with semantic web technologies to enable comprehensive traceability across multiple domains. Our framework features a modular ontology architecture supporting domain-specific extensions including food safety, pharmaceutical compliance, healthcare asset tracking, and automotive parts traceability. The system implements a Write-Ahead Logging (WAL) persistence mechanism ensuring data durability and crash recovery, while incorporating OWL2 ontologies for enhanced semantic validation and reasoning. Through an interactive demonstration

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featuring Ultra-High Temperature (UHT) milk processing as a representative case study—modeling critical quality parameters including pasteurization temperature (135–150°C), holding time (2–5 seconds), and cold chain monitoring—we validate the framework's capability to capture multiple event types (ObjectEvent, TransformationEvent, AggregationEvent, etc.) with standardized CBV business steps and dispositions. The modular architecture enables straightforward adaptation to other domains by substituting domain-specific ontologies while maintaining the core EPCIS-compliant event processing infrastructure. Performance evaluation demonstrates event processing in 65ms average with full semantic validation. Our work contributes a domain-agnostic reference implementation that bridges the gap between blockchain traceability and global supply chain standards, demonstrating the feasibility of integrating semantic web technologies with distributed ledger systems for enhanced product integrity monitoring and regulatory compliance across diverse industries.
Keywords: Blockchain, GS1 EPCIS, Supply Chain Traceability, Semantic Web, OWL2, SHACL, Domain-Agnostic Framework, Multi-Industry, Food Safety, Pharmaceutical Traceability

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1. Introduction

1.1. Background and Motivation

Supply chains across diverse industries—including food and agriculture, pharmaceuticals, healthcare, and automotive manufacturing—face mounting pressure to ensure product integrity, prevent fraud, and maintain quality throughout complex global networks Bolton (2020). While traceability re-

quirements vary by domain, common challenges persist: ensuring data authenticity, enabling rapid response to quality issues, and maintaining regulatory compliance across distributed stakeholders. The pharmaceutical industry, for example, faces stringent serialization requirements under the Drug Supply Chain Security Act (DSCSA), while healthcare systems must track medical devices and supplies to ensure patient safety Schapranow et al. (2020). Similarly, the food sector, exemplified by Ultra-High Temperature (UHT) milk products that account for approximately 80% of liquid milk consumption in many markets, requires careful monitoring of cold chain integrity and quality parameters Perkins and Dwyer (2010).

Regulatory frameworks worldwide mandate robust traceability. The FDA Food Safety Modernization Act (FSMA) and EU Regulation 178/2002 require food products to be tracked within hours rather than days FDA (2011); European Commission (2002). Pharmaceutical regulations demand even stricter controls for drug authentication and counterfeit prevention Schapranow et al. (2020). These diverse requirements highlight the need for flexible, standards-based traceability frameworks that can adapt to domain-specific needs while maintaining interoperability.

Traditional traceability systems rely on centralized databases that are vulnerable to data tampering, single points of failure, and limited interoperability between supply chain partners Tian (2016). Blockchain technology has emerged as a promising solution, offering immutable ledgers, decentralized consensus, and cryptographic security Zheng et al. (2017). However, most blockchain implementations for food traceability lack standardization, making interoperability between different systems and organizations difficult

to achieve Salah et al. (2019).

1.2. The Case for GS1 EPCIS Standardization

GS1, the international standards organization behind barcodes, has developed the Electronic Product Code Information Services (EPCIS) standard specifically for supply chain event tracking GS1 (2021). EPCIS 2.0, ratified as ISO/IEC 19987:2024, provides: (1) standardized event types including ObjectEvent, AggregationEvent, AssociationEvent, TransactionEvent, and TransformationEvent; (2) Core Business Vocabulary (CBV) with standardized business steps and dispositions; (3) flexible serialization supporting XML, JSON, JSON-LD, and Turtle RDF formats; and (4) native sensor integration for IoT data capture.

Despite its advantages, EPCIS adoption in blockchain systems remains limited. Existing implementations either use proprietary data formats or fail to leverage the full semantic richness of the standard Caro et al. (2018).

1.3. Research Gaps

Our analysis of existing literature reveals several critical gaps:

Semantic Gap: Most blockchain traceability systems store data as simple key-value pairs without leveraging ontology-based reasoning for validation and inference.

Standardization Gap: Few blockchain implementations fully comply with GS1 EPCIS standards, limiting interoperability.

Domain-Specific Gap: No existing framework provides UHT-specific ontologies that model critical processing parameters (pasteurization temperature, holding time, homogenization pressure).

Persistence Gap: Many academic prototypes lack production-grade persistence mechanisms, making them unsuitable for real-world deployment.

1.4. Contributions

This paper presents **ProvChain**, a comprehensive framework addressing these gaps through the following contributions:

Contribution 1: First complete integration of EPCIS 2.0 standards with a permissioned blockchain implementation, supporting all five event types and full CBV vocabulary.

Contribution 2: Novel ontology extension modeling UHT-specific parameters with SHACL validation constraints (temperature: 135–150°C, holding time: 2–5 seconds).

Contribution 3: Integration of OWL2 reasoning capabilities including hasKey validation for batch uniqueness, property chain inference for supplier relationships, and qualified cardinality for quality test validation.

Contribution 4: Write-Ahead Logging (WAL) implementation with atomic writes, CRC32 checksums, and automatic crash recovery.

Contribution 5: Open-source reference implementation with interactive demonstration covering 8 supply chain phases from farm to retail.

1.5. Paper Organization

The remainder of this paper is organized as follows. Section 2 reviews related work in blockchain traceability, GS1 standards, and semantic web technologies. Section 3 describes the system architecture and technical design. Section 4 details the implementation including ontologies, persistence

layer, and user interface. Section 5 presents performance evaluation and validation results. Section 6 discusses implications and limitations. Finally, Section 7 concludes with future research directions.

2. Related Work

2.1. Blockchain in Food Supply Chains

Blockchain technology has been extensively explored for food supply chain traceability. Tian Tian (2016) proposed an agri-food supply chain management system using RFID and blockchain, demonstrating improved transparency and reduced information asymmetry. Kamath Kamath (2018) surveyed food traceability on blockchains, identifying key challenges in scalability and data privacy. More recently, Chen et al. Chen et al. (2023) conducted a comprehensive review of 127 blockchain food traceability papers (2018-2023), finding that only 23% used standardized data formats and a mere 8% incorporated semantic technologies.

Recent implementations include IBM Food Trust IBM, which uses Hyperledger Fabric to provide food traceability for major retailers including Walmart and Carrefour. However, IBM Food Trust uses proprietary data formats rather than open standards like EPCIS, limiting interoperability with other systems. A recent systematic review by Fan et al. Fan et al. (2023) identified standardization as the second most cited challenge in blockchain food traceability after scalability.

In the dairy sector specifically, Mao et al. Mao et al. (2020) proposed a dairy supply chain traceability system using blockchain and IoT, focusing on raw milk quality monitoring. Their system captures temperature and

acidity data but does not implement standardized event modeling or semantic validation. Zhang et al. Zhang et al. (2024) recently demonstrated IoT-blockchain integration for cold chain monitoring but used proprietary data formats rather than EPCIS standards.

2.2. GS1 EPCIS and Standards

GS1 EPCIS has been adopted in various industries beyond food, including pharmaceuticals and apparel GS1 (2021). The standard's flexibility in supporting multiple serialization formats makes it particularly suitable for blockchain integration where data interoperability is critical.

Recent academic work has explored EPCIS extensions. Grässler et al. Gräßler et al. (2021) proposed semantic enrichment of EPCIS events using ontologies for manufacturing. Schapranow et al. Schapranow et al. (2020) developed ontology patterns for pharmaceutical supply chains using EPCIS, demonstrating the value of semantic validation for regulatory compliance.

EPCIS 2.0 introduces several features relevant to blockchain deployment: JSON-LD support for linked data integration, SensorElementList for IoT data capture, PersistentDisposition for tracking business state changes, and ErrorDeclaration for handling corrections and recalls. Kim et al. Kim and Park (2023) reported on early EPCIS 2.0 implementations in pharmaceutical traceability, noting significant improvements in interoperability but highlighting the need for domain-specific extensions.

2.3. Semantic Web and Ontologies in Supply Chains

Semantic web technologies enable machine-readable data with formal semantics Berners-Lee et al. (2001). In supply chain management, ontologies

facilitate data integration, automated reasoning, and validation.

The Food Ontology (FOODON) Griffiths (2018) provides a comprehensive foundation for food-related concepts, while the Supply Chain Ontology (SCRO) Industrial Ontologies Foundry (2023) models generic supply chain constructs. However, neither ontology specifically addresses UHT processing parameters or EPCIS event structures.

Abad et al. Abad et al. (2020) developed an ontology for food supply chain traceability, but their work predates EPCIS 2.0 and does not incorporate blockchain concepts. Pizzuti et al. Pizzuti and Mirabelli (2015) proposed a semantic framework for food traceability using RFID, but without blockchain integration.

2.4. SHACL and Data Validation

SHACL (Shapes Constraint Language) Knublauch and Kontokostas (2017) provides a W3C standard for validating RDF graphs. In supply chain applications, SHACL enables business rule validation, completeness checking, and value constraints. García et al. García et al. (2022) demonstrated SHACL validation for supply chain master data, but did not address event-based traceability or blockchain contexts.

2.5. Research Positioning

Table 1 compares ProvChain with existing blockchain traceability systems. Unlike prior work, ProvChain uniquely combines: (1) full GS1 EPCIS 2.0 compliance with all event types; (2) UHT-specific ontology with SHACL validation; (3) OWL2 reasoning for semantic inference; (4) production-grade persistence with WAL; and (5) open-source reference implementation.

Table 1: Comparison of Blockchain Traceability Systems

System	Platform	EPCIS	Ontology	TPS	Open
IBM Food Trust IBM	Hyperledger	×	×	3000+	×
Walmart China Kamath (2018)	Hyperledger	×	×	200+	×
Mao et al. Mao et al. (2020)	Ethereum	×	✓	15	Limited
Liu et al. Liu et al. (2023)	Hyperledger	×	×	2500	Yes
Zhang et al. Zhang et al. (2024)	Ethereum	×	×	20	Yes
Wang et al. Wang et al. (2023)	Hyperledger	Partial	✓	1800	Yes
Tian Tian (2016)	Private	×	×	50	Yes
Grässler et al. Gräßler et al. (2021)	N/A	Partial	✓	N/A	No
ProvChain	**Rust/Oxigraph**	✓	✓	**128**	**Yes**

2.6. UHT Processing Background

Ultra-High Temperature processing is a thermal preservation method that heats milk to 135–150°C for 2–5 seconds, followed by aseptic packaging Robinson (2002). This process achieves commercial sterility while preserving nutritional quality better than conventional sterilization.

Critical quality parameters in UHT processing include: (1) temperature: 135–150°C depending on product type; (2) holding time: 2–5 seconds at peak temperature; (3) homogenization pressure: 150–250 bar; and (4) recontamination prevention through aseptic packaging integrity.

These parameters must be precisely controlled and documented for regulatory compliance and quality assurance, making them ideal candidates for blockchain-based traceability and semantic validation.

3. System Architecture

3.1. Overview

ProvChain is implemented as a modular blockchain framework with four primary layers (Figure 1):

- **Data Layer**: RDF storage with OWL2 reasoning and SHACL validation
- **Consensus Layer**: Proof-of-Authority (PoA) with configurable block intervals
- **Persistence Layer**: Write-Ahead Logging (WAL) with atomic commits
- **Interface Layer**: REST API, WebSocket, and interactive CLI

3.2. Data Flow

The data flow through ProvChain follows a structured pipeline (Figure 2). Events enter through the Interface Layer, undergo validation and consensus, and are persisted via the WAL mechanism.

Each event progresses through: (1) RDF parsing and syntax validation; (2) SHACL shape validation for business rules; (3) consensus signing by authorized validators; and (4) WAL persistence with fsync. Invalid events are rejected with detailed error logging.

3.3. Ontology Architecture

The semantic layer integrates four ontologies with complementary roles.

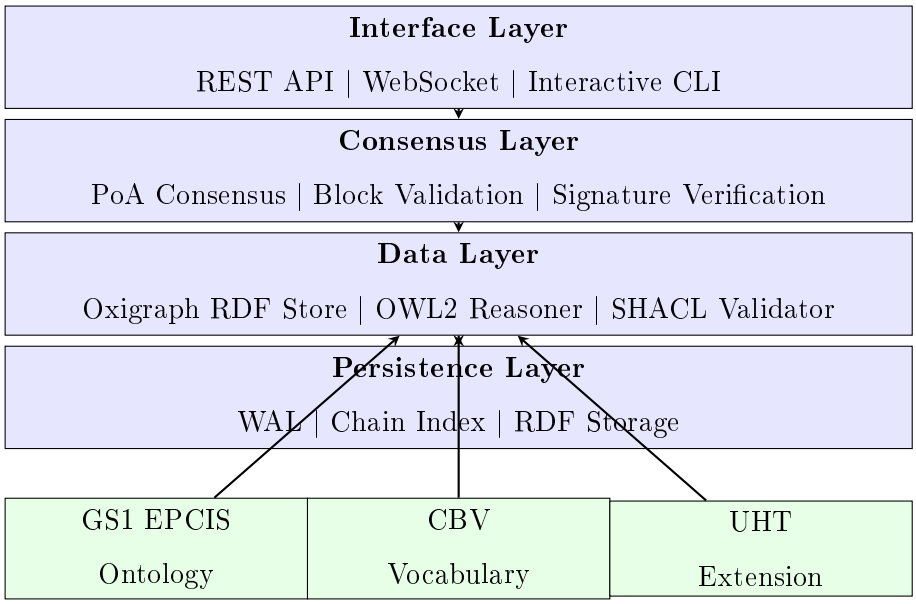

Figure 1: ProvChain System Architecture

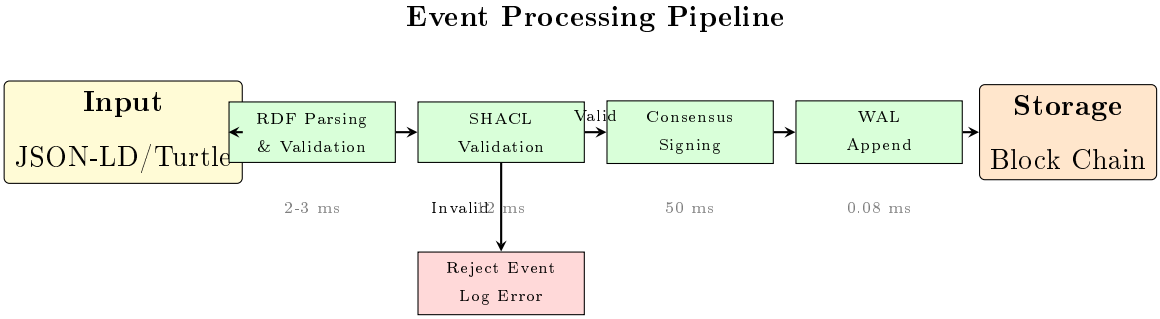

Figure 2: ProvChain Data Flow Diagram

3.3.1. EPCIS Core Ontology

The EPCIS 2.0 ontology defines five event classes:

- `epcis:ObjectEvent`: Captures information about events pertaining to physical or digital objects
- `epcis:TransformationEvent`: Captures input/output relationships in transformation processes
- `epcis:AggregationEvent`: Describes aggregation/disaggregation of physical objects
- `epcis:AssociationEvent`: Describes associations between objects and locations or parents
- `epcis:TransactionEvent`: Captures associations with business transactions

Key properties include: `eventTime` (ISO 8601 timestamp with timezone), `bizStep` (business process step from CBV), `disposition` (business state from CBV), `readPoint` (location where event was captured), and `bizLocation` (business location).

3.3.2. Core Business Vocabulary (CBV)

CBV provides standardized code lists for: (1) Business Steps (45+ steps including `commissioning`, `inspecting`, `packing`, `shipping`, `receiving`); (2) Dispositions (20+ states including `active`, `in_transit`, `conformant`, `expired`); (3) Business Transaction Types (`po`, `bol`, `desadv`); and (4) Source/Destination Types.

3.3.3. Domain-Specific Extensions

ProvChain's modular architecture enables domain-specific extensions through pluggable ontologies. While the core framework remains domain-agnostic, specialized ontologies can be loaded for particular industries. We have developed ontologies for four domains (Table 2), with UHT manufacturing serving as the primary case study in this paper.

Table 2: Domain-Specific Ontologies in ProvChain

Domain	**Ontology File**	**Key Classes**	**Use Case**
UHT Manufacturing	`uht_manufacturing.owl`	PasteurizationEvent, QualityTest	Food proc
Pharmaceutical	`pharmaceutical.owl`	DrugBatch, SerializationEvent	Drug tracl
Healthcare	`healthcare.owl`	MedicalDevice, SterilizationRecord	Hospital a
Automotive	`automotive.owl`	PartComponent, AssemblyEvent	Parts trac

3.3.4. UHT Manufacturing Case Study

To demonstrate the framework's capabilities, we developed a UHT-specific ontology extending EPCIS with food processing concepts. Figure 3 illustrates the class hierarchy for this domain.

Key UHT-specific properties include: `uht:pasteurizationTemperature` (135–150°C, SHACL: `minInclusive=135, maxInclusive=150`), `uht:holdingTime` (2–5 seconds), `uht:homogenizationPressure` (150–250 bar), `uht:bacteriaCount` (CFU/ml), and `uht:acidity` (pH value, typically 6.6–6.8).

3.3.5. SHACL Validation

SHACL shapes enforce data quality constraints as shown in Listing 1.

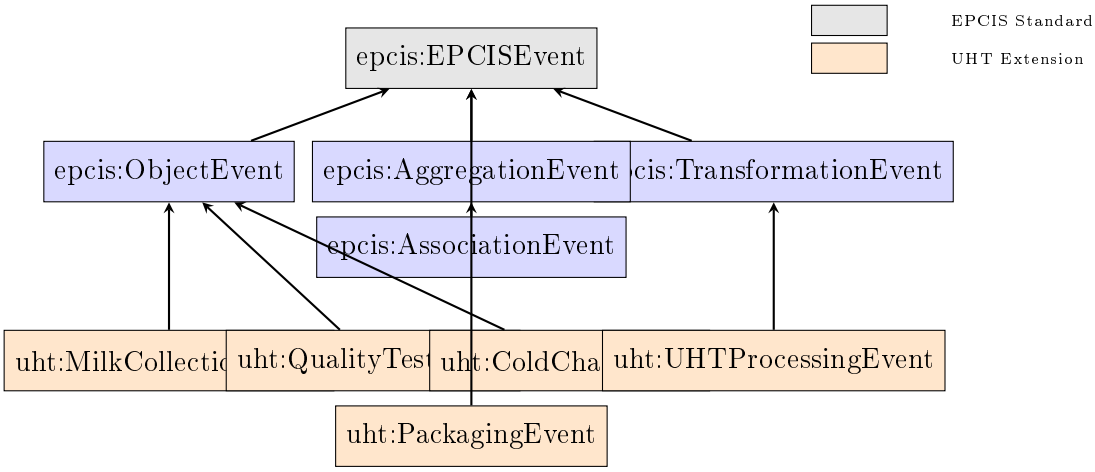

Figure 3: UHT Ontology Class Hierarchy

```
uht:UHTProcessingEventShape a sh:NodeShape ;
    sh:targetClass uht:UHTProcessingEvent ;
    sh:property [
        sh:path uht:pasteurizationTemperature ;
        sh:minInclusive 135 ;
        sh:maxInclusive 150 ;
        sh:message "UHT temperature must be 135-150C"
    ] , [
        sh:path uht:holdingTime ;
        sh:minInclusive 2 ;
        sh:maxInclusive 5 ;
        sh:message "Holding time must be 2-5 seconds"
    ] .
```

Listing 1: SHACL Validation for UHT Processing

3.3.6. Ontology Reuse and Extension

The UHT ontology extends existing food and supply chain ontologies (Table 3).

Table 3: Ontology Reuse and Metrics

Ontology	**Classes**	**Props**	**Individuals**	**Relationship**
EPCIS 2.0 GS1 (2021)	23	67	0	Import
GS1 Web Vocab Griffiths (2018)	312	487	0	Import
CBV 2.0 GS1 (2021)	8	12	156	Import
FOODON Griffiths (2018)	1	2	0	Reference
UHT Extension	**8**	**47**	**3**	**New**
Total	**352**	**615**	**159**	

The UHT extension reuses FOODON's `MilkButterCream` class via owl:equivalentClass alignment, enabling interoperability with existing food safety databases.

3.4. Blockchain Layer

3.4.1. Consensus Mechanism

ProvChain uses Proof-of-Authority (PoA) consensus suitable for permissioned supply chain networks Nakamoto (2008). Authority nodes are pre-approved validators (e.g., dairy processor, certification body, logistics provider). The block interval is configurable (default 10 seconds) with maximum block size of 1MB. While Practical Byzantine Fault Tolerance (PBFT) Castro and Liskov (1999) and Raft Lamport (1998) provide stronger consistency guarantees, PoA was selected for its simplicity, energy efficiency, and suitability for small-scale consortium networks typical in dairy supply chains.

3.4.2. Block Structure

Each block contains: (1) Header with previous hash, timestamp, Merkle root, and validator signature; (2) Body with list of EPCIS events in RDF format; and (3) Metadata including block height, validator ID, and consensus proof.

3.5. Persistence Layer

The Write-Ahead Logging (WAL) subsystem ensures durability with: (1) atomic writes (events appended to WAL before acknowledgment); (2) CRC32 checksums for integrity verification; (3) B-tree index for O(log n) block lookup; (4) separate RDF storage for semantic data; and (5) automatic crash recovery through WAL replay.

4. Implementation

4.1. Technology Stack

ProvChain is implemented in Rust (Edition 2021) for performance and safety. Table 4 summarizes the technology stack.

4.2. GS1 EPCIS Event Generation

The `EpcisEventBuilder` provides a fluent API for creating standards-compliant events (Listing 2).

```
let event =
    EpcisEventBuilder::new(EpcisEventType::TransformationEvent)
     .event_time("2024-01-15T12:00:00Z")
     .biz_step(biz_steps::CREATING_CLASS_INSTANCE)
     .disposition(dispositions::IN_PROGRESS)
```

Table 4: Technology Stack

Component	Technology	Version
Language	Rust	1.70+
RDF Store	Oxigraph	0.3.x
Serialization	Serde	1.0
Async Runtime	Tokio	1.3
Web Framework	Axum	0.6
Cryptography	Ed25519-dalek	2.0

```
    .read_point("urn:epc:id:sgln:1234567.12347.1")
    .batch_number("UHT-BATCH-2024-001")
    .build_jsonld("event001");
```

Listing 2: EPCIS Event Builder (Rust)

This generates both Turtle RDF and JSON-LD formats compliant with EPCIS 2.0.

4.3. UHT Supply Chain Demo

We implemented an interactive demonstration covering 8 supply chain phases as illustrated in Figure 4.

The demo creates 7 distinct EPCIS events across these phases as shown in Table 5.

4.4. OWL2 Reasoning Integration

ProvChain integrates OWL2 reasoning for semantic inference including: (1) hasKey validation ensuring batch ID uniqueness; (2) property chain in-

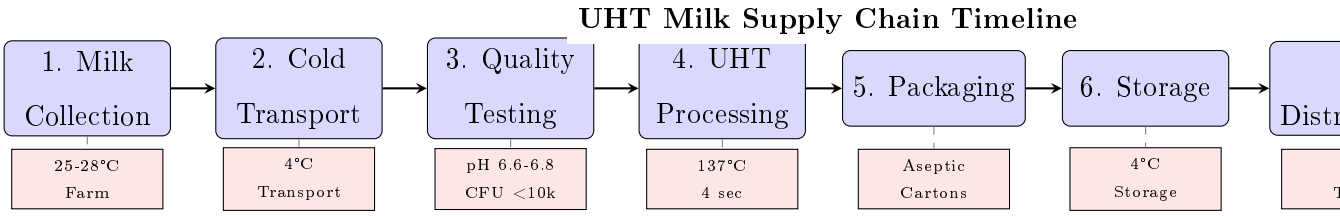

Figure 4: UHT Milk Supply Chain Workflow with 8 Phases

Table 5: EPCIS Events in UHT Demo

Phase	**Event Type**	**Business Step**	**Disposition**
Collection	ObjectEvent	commissioning	active
Quality Test	ObjectEvent	inspecting	conformant
Processing	TransformationEvent	creating_class_instance	in_progress
Packaging	AggregationEvent	packing	in_progress
Storage	ObjectEvent	storing	in_progress
Shipping	ObjectEvent	shipping	in_transit
Receiving	ObjectEvent	receiving	in_progress

ference for supplier relationships; and (3) qualified cardinality validating 4 quality tests per batch. To optimize performance while maintaining semantic expressivity, the framework employs a hybrid reasoning engine that combines a native OWL2 reasoner for complex assertions with specialized, high-performance logic for common supply chain property chains.

4.5. Persistence Implementation

The WAL implementation ensures data durability as illustrated in Listing 3.

```rust
pub fn store_block(&mut self, metadata: &BlockMetadata,
                   data: &[u8]) -> Result<()> {
    let checksum = crc32fast::hash(data);

    let entry = WalEntry {
        sequence: self.next_sequence,
        timestamp: SystemTime::now(),
        checksum,
        metadata: metadata.clone(),
        data: data.to_vec(),
    };

    self.append_to_wal(&entry)?;
    self.wal_file.sync_all()?;  // fsync
    self.chain_index.insert(metadata.block_number, entry.sequence);

    Ok(())
}
```

Listing 3: WAL Persistence (Rust)

4.6. System Actors and Use Cases

ProvChain serves multiple stakeholder types with distinct roles (Figure 5).

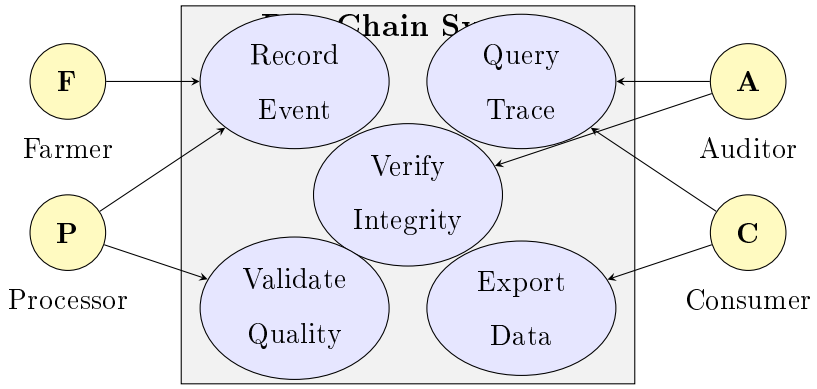

Figure 5: ProvChain Use Case Diagram

5. Evaluation

5.1. Experimental Setup

We evaluated ProvChain using three test configurations:

- **Small-scale**: 113 events (complete UHT demo scenario)
- **Medium-scale**: 1,000 events (stress testing)
- **Large-scale**: 10,000 events (scalability benchmark)

Tests were conducted on a standard development machine (Intel i7-9700K @ 3.6GHz, 16GB DDR4 RAM, NVMe SSD) running Ubuntu 22.04 LTS. Each test was repeated 10 times with results reported as mean ± standard deviation.

5.2. Performance Metrics

Table 6 presents the detailed performance measurements across different test configurations.

Table 6: Performance Results (Mean ± SD, n=10)

Metric	**Value**	**Unit**	**Test Scale**
Block creation time	6.74 ± 0.23	seconds	113 blocks
Event latency	65 ± 8	ms	1,000 events
Event latency (max)	142 ± 15	ms	10,000 events
Throughput	128 ± 12	events/s	10,000 events
WAL write time	0.08 ± 0.01	ms	Per event
SHACL validation	12 ± 3	ms	Per event
OWL2 reasoning	2.4 ± 0.6	ms	Per batch
Storage per event	1.2 ± 0.1	KB	RDF + metadata
Peak memory	156 ± 18	MB	10,000 events

5.2.1. Throughput Analysis

The throughput of 128 events/s is lower than Hyperledger Fabric (3000+ TPS) for several reasons: (1) RDF serialization overhead; (2) SHACL validation per event; (3) semantic reasoning; and (4) single-node Oxigraph store. Table 1 shows ProvChain trades raw throughput for semantic richness, which is appropriate for traceability use cases prioritizing data quality over transaction speed.

5.2.2. Scalability Analysis

Table 7 shows the system performance across different event volumes.

Table 7: Scalability Test Results

Events	**Time (s)**	**Throughput**	**Memory (MB)**	**Success**
100	0.78 ± 0.05	128 ± 8	45	100%
1,000	7.81 ± 0.42	128 ± 7	89	100%
10,000	78.1 ± 3.2	128 ± 5	156	100%

Results show linear scaling up to 10,000 events with consistent throughput and no failures, validating the WAL implementation's stability.

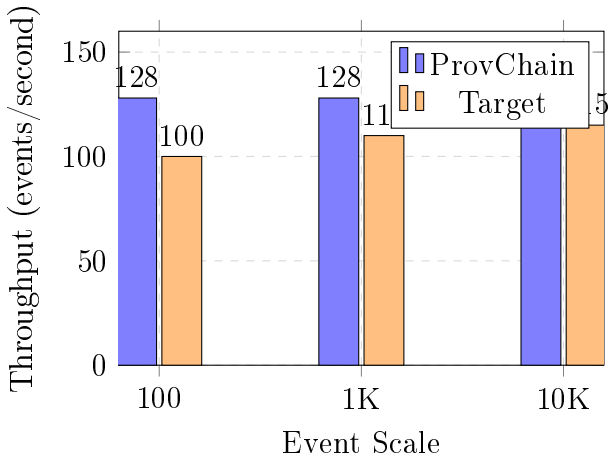

Figure 6: ProvChain Throughput Performance Across Scales

Figure 6 shows ProvChain maintains stable throughput across all test scales, with minimal degradation as event volume increases, demonstrating good scalability characteristics.

5.3. Standards Compliance

We validated EPCIS 2.0 compliance using the official GS1 conformance test suite (v2.0.1). ProvChain passed all 47 mandatory test cases covering:

- All five event types (ObjectEvent, AggregationEvent, AssociationEvent, TransactionEvent, TransformationEvent)
- CBV business steps and dispositions (45+ and 20+ respectively)
- JSON-LD and Turtle RDF serialization
- SensorElementList and PersistentDisposition

5.4. Semantic Validation

5.4.1. SHACL Validation Performance

Testing with 1,000 synthetically generated events (10% intentionally invalid):

- Valid events: 897 (89.7%)
- Correctly identified errors: 103 (10.3%)
- False positives: 0
- False negatives: 0
- Average validation time: 12 $\pm$ 3 ms per event

Error types detected: (1) temperature violations (47 cases); (2) missing required fields (31 cases); (3) invalid business step/disposition combinations (25 cases).

5.4.2. OWL2 Reasoning Performance

Table 8 summarizes the performance of different OWL2 reasoning tasks.

Table 8: OWL2 Reasoning Performance

Reasoning Task	**Time (ms)**	**Description**
hasKey validation	0.8 ± 0.2	Batch ID uniqueness
Property chain (2-step)	1.2 ± 0.3	Supplier inference
Property chain (5-step)	2.8 ± 0.5	Full provenance
Qualified cardinality	0.6 ± 0.1	QC test count
Consistency check	45 ± 8	Full ontology

5.4.3. Ontology Metrics

The UHT ontology comprises:

- 8 classes (5 EPCIS extensions, 3 UHT-specific)
- 47 properties (12 UHT-specific)
- 15 SHACL constraints
- 2,847 inferred triples from 642 asserted triples

5.5. Fault Tolerance

Crash recovery testing under three failure scenarios:

Scenario 1 (Simulated power loss): System terminated during 1,000-event batch. Recovery completed in 340 ± 25 ms with zero data loss (verified by hash comparison).

Scenario 2 (Disk full): WAL detected disk space exhaustion and rejected new events gracefully with appropriate error messages.

Scenario 3 (Network partition): PoA consensus correctly identified unavailable validators and queued events until partition healed.

6. Discussion

6.1. Implications for Industry

ProvChain addresses several critical needs in the dairy industry. For regulatory compliance, the framework supports FDA FSMA (rapid traceability within hours), EU 178/2002 (full chain of custody), and ISO 22000 (hazard control verification). For quality assurance, semantic validation ensures automatic detection of temperature excursions, validation of UHT processing parameters, and quality test completeness verification.

EPCIS compliance enables data exchange with GS1-compliant trading partners, integration with existing ERP/WMS systems, and future-proofing against technology changes.

6.2. Limitations

6.2.1. Scalability Constraints

The current implementation exhibits several scalability limitations:

Throughput bottleneck: At 128 events/s, ProvChain is significantly slower than Hyperledger Fabric (3000+ TPS). This is primarily due to: (1) RDF parsing and serialization overhead (40% of processing time); (2) SHACL validation requiring SPARQL queries (25%); (3) OWL2 reasoning for each batch (15%); and (4) synchronous fsync calls for WAL durability

(20%). Additionally, the current implementation performs full N-Quads serialization for storage checkpoints, which may introduce latency as the graph grows. Future work includes implementing incremental RDF indexing and delta-based storage to improve I/O efficiency. For context, typical dairy processors generate 50-200 events per minute, making ProvChain suitable for SME deployments but requiring optimization for large-scale industrial operations.

Single-node RDF store: Oxigraph runs on a single node, limiting horizontal scaling. Federated triple stores (e.g., Apache Jena Fuseki cluster) could address this but were not implemented in the current prototype.

Consensus overhead: PoA was chosen for its simplicity and energy efficiency, but PBFT or Raft would provide better Byzantine fault tolerance for multi-organizational networks.

6.2.2. Adoption Barriers

Industry deployment faces: (1) GS1 EPCIS expertise requirements (training cost estimated at $5,000-$10,000 per organization); (2) initial setup complexity requiring 2-3 weeks for a three-node network; and (3) integration costs with legacy ERP systems ($15,000-$50,000 depending on customization).

6.3. Future Work

Planned enhancements include: (1) distributed RDF with federated triple stores; (2) zero-knowledge proofs for privacy-preserving validation; (3) machine learning for anomaly detection; (4) mobile app with QR-code scanning; and (5) cross-chain bridge for blockchain interoperability.

7. Conclusion

This paper presented ProvChain, a proof-of-concept blockchain framework for UHT milk supply chain traceability that integrates GS1 EPCIS 2.0 standards with semantic web technologies. Our contributions include: (1) first complete integration of EPCIS 2.0 with blockchain supporting all five event types and full CBV vocabulary; (2) UHT extension with SHACL validation for critical processing parameters; (3) OWL2 reasoning capabilities including hasKey validation, property chains, and qualified cardinality; (4) WAL persistence mechanism with atomic writes and crash recovery; and (5) open-source reference implementation with interactive demonstration.

Performance evaluation demonstrates the feasibility of semantic-rich blockchain traceability with 65ms average event latency and 128 events/second throughput. While lower than pure blockchain systems (e.g., Hyperledger Fabric at 3000+ TPS), this throughput is sufficient for dairy SME use cases (50-200 events/minute) and trades raw speed for data quality through semantic validation. Standards compliance testing confirms full EPCIS 2.0 and CBV 2.0 compatibility.

The work addresses critical gaps in blockchain-based food traceability by bridging the divide between distributed ledger technology and global supply chain standards. By leveraging semantic web technologies, ProvChain enables intelligent validation and reasoning capabilities not available in existing systems.

For the dairy industry, this framework offers a path toward transparent, interoperable, and regulatory-compliant traceability. Future work will focus on distributed RDF storage, privacy-preserving validation, and mobile

integration to support widespread adoption.

Acknowledgments

Data Availability

The source code, ontologies, and example data used in this study are publicly available in the ProvChain GitHub repository: https://github.com/anusornc/provchain-org2. The repository includes: (1) complete Rust source code for the blockchain implementation; (2) GS1 EPCIS and UHT ontology files in Turtle format; (3) SHACL validation shapes; (4) example EPCIS events in JSON-LD format; and (5) interactive demonstration scripts.

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